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Memristive control of mutual spin Hall nano-oscillator synchronization for neuromorphic computing

Mohammad Zahedinejad[1,2], Himanshu Fulara[1,3], Roman Khymyn[1], Afshin Houshang[1], Mykola Dvornik[2], Shunsuke Fukami[4,5,6,7,8], Shun Kanai[4,5,6,9], Hideo Ohno[4,5,6,7,8] and Johan Åkerman[1,2,10] ✉

Synchronization of large spin Hall nano-oscillator (SHNO) arrays is an appealing approach toward ultrafast non-conventional computing. However, interfacing to the array, tuning its individual oscillators and providing built-in memory units remain substantial challenges. Here, we address these challenges using memristive gating of W/CoFeB/MgO/AlO$_x$-based SHNOs. In its high resistance state, the memristor modulates the perpendicular magnetic anisotropy at the CoFeB/MgO interface by the applied electric field. In its low resistance state the memristor adds or subtracts current to the SHNO drive. Both electric field and current control affect the SHNO auto-oscillation mode and frequency, allowing us to reversibly turn on/off mutual synchronization in chains of four SHNOs. We also demonstrate that two individually controlled memristors can be used to tune a four-SHNO chain into differently synchronized states. Memristor gating is therefore an efficient approach to input, tune and store the state of SHNO arrays for non-conventional computing models.

The demand for compact communication systems and faster processing hardware is booming as we approach the era of the Internet of Things. However, with the approaching end of Moore's law and the increasing demand for mobile applications, it seems unlikely that state-of-the-art complementary metal-oxide-semiconductor (CMOS) technologies can live up to all Internet of Things expectations[1]. Emerging models—such as machine vision and cognitive tasks for connected objects—have severely challenged the existing technologies in terms of power consumption and processing speed, tasks that the human brain, thanks to its massive neuronal connectivity, performs extremely fast using negligible power compared to conventional processors[2]. Although there have been efforts to imitate neuronal connectivity by involving many transistors representing individual neurons, von Neumann computing inherently does not include connected and interactive computing elements[3]. Numerous types of unconventional computing hardware have been proposed by different research communities to address such von Neumann shortcomings, each outperforming CMOS processors in some aspects: Memristive computing[4,5], optical computing[6,7], spintronic computing[8–10] and quantum computing[11] are only a few examples of such hardware.

Using coupled networks of oscillators as an alternative computing model has been proposed since many different technologies can explore this route, ranging from mechanical oscillators[12] through memristive[13,14] and superconducting oscillators[6] to optical[15] and spintronic[8,16–21] oscillators. Among these, the spintronic oscillator is a promising candidate meeting all technical requirements, such as room temperature operation, true miniaturization, CMOS integration, high-speed and low power consumption. In a recent study, a short chain of four weakly coupled spin transfer torque nano-oscillators (STNOs) was used to perform vowel recognition[17]. Through tuning of the individual STNO currents, and their injection phase-locking to two external microwave sources, a recognition rate comparable to state-of-the-art CMOS processors was achieved. However, more complex tasks will require many more oscillators, and the demonstrated STNOs do not offer an attractive scaling path to much larger networks. We have recently demonstrated two-dimensional (2D) synchronization in large arrays of an alternative emerging class of nanoscale spin-based oscillators—so-called SHNOs[22]. Arrays accommodating up to 64 strongly coupled SHNOs, operating at 10 GHz, achieved robust synchronization and a corresponding record-high quality factor of 170,000. We also demonstrated that the 2D-SHNO arrays were capable of performing the same type of computation as STNO chains[17] while they operate at a 25 times higher frequency. Adding the potential to control individual SHNOs using direct voltage control[23], such SHNO arrays offer a viable scaling path to very large nano-oscillator networks of SHNOs that can be scaled down to 20 nm (refs. [24,25]). In addition, most oscillator network computation proposals require not only interacting oscillators but also on-chip memory for training purposes[12,26–29]. Hence there is both a practical and computational need for non-volatile storage of individual oscillator properties: a challenge that remains unsolved for nanoscale arrays.

Metal-oxide memristive devices based on metal insulator metal materials are one of the promising candidates for scalable and dense memory, and in-memory processing applications. On increasing

[1]Physics Department, University of Gothenburg, Gothenburg, Sweden. [2]NanOsc AB, Kista, Sweden. [3]Department of Physics, Indian Institute of Technology Roorkee, Roorkee, India. [4]Laboratory for Nanoelectronics and Spintronics, Research Institute of Electrical Communication, Tohoku University, Sendai, Japan. [5]Center for Science and Innovation in Spintronics, Tohoku University, Sendai, Japan. [6]Center for Spintronics Research Network, Tohoku University, Sendai, Japan. [7]Center for Innovative Integrated Electronic Systems, Tohoku University, Sendai, Japan. [8]WPI-Advanced Institute for Materials Research, Tohoku University, Sendai, Japan. [9]Division for the Establishment of Frontier Sciences, Tohoku University, Sendai, Japan. [10]Material and Nanophysics, School of Engineering Sciences, KTH Royal Institute of Technology, Kista, Sweden. ✉e-mail: johan.akerman@physics.gu.se

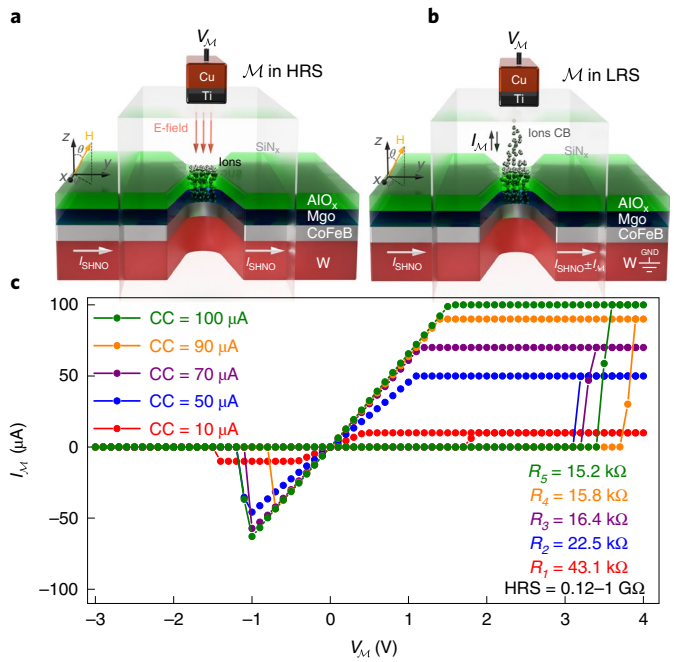

Fig. 1 | Device schematic and multi-level resistive switching of the memristor gate. a,**b**, Schematic of a SHNO with a memristive gate fabricated on top of the nano-constriction sharing a common ground contact when operating in HRS (**a**) and LRS. A W(5)/$(Co_{0.75}Fe_{0.25})_{75}B_{25}$(1.7)/MgO(2)/$AlO_x$(2) (**b**) multilayer (numbers in parentheses indicate thickness in nm) is buried in SiN_x(15) with a Ti(2)/Cu(40) top electrode as gate contact. The coordinate system shows the magnetic field vector $\mu_0 H$ applied at an out-of-plane angle θ and an in-plane angle ϕ. On sweeping $V_{\mathcal{M}}$, an electric field-driven reversible conductive bridge is formed and retracted through the insulating multilayer and the memristor current $I_{\mathcal{M}}$ is added or subtracted from I_{SHNO} depending on the $V_{\mathcal{M}}$ polarity. **c**, Multi-level resistive switching of the memristive gate, controlled by the current compliance of the power supply.

the applied electric field (voltage) across the metal oxide in its pristine, insulating, high resistance state (HRS), depending on the choice of electrodes, a conductive ionic bridge starts to form by the migration of oxygen anions or metal cations species[30]. Once the ionic bridge touches both electrodes, it transforms the insulator to a locally conductive layer, known as low resistance state (LRS). In the LRS, if the applied voltage is removed, the conductive ionic bridge remains intact, hence the resistance state is non-volatile. By controlling the radial size of the conductive bridge, several LRSs can be realized, therefore, memristors are ideal candidates for multi-states memory elements.

In this work we present an ultra-low-power path to controlling the magnetization dynamics in W/$(Co_{0.75}Fe_{0.25})_{75}B_{25}$–/MgO-based SHNOs by applying an electric field via gating. The use of electric field (voltage) to control magnetism has been extensively studied in the literature, for example, using voltage controlled magnetic anisotropy (VCMA) to achieve low-power switching in magnetic tunnel junction memory cells by controlling or modulating the perpendicular magnetic anisotropy (PMA)[31–35]. As PMA can have a substantial impact on the operating frequency and the auto-oscillation (AO) mode profile of W/CoFeB/MgO-based SHNOs[36,37], our voltage control of the CoFeB–MgO interface tunes the local magnetodynamical properties, resulting in a change in the precessional angle and the mode volume[23,38] that define the SHNO frequency and the coupling between SHNOs. In addition to the direct voltage control, we also show that the gates can exhibit a pronounced memristive behaviour, which we then use to control individual oscillators in a chain to make them either operate in synchrony or break their synchronization into different states of partial synchronization. Synchronization control can be achieved in both the HRS and the LRS of the memristor, albeit through different mechanisms: electric field-driven in the HRS and current-driven in the LRS. The memristors also add embedded memory functionality that can be used to recall the past input values each oscillator was exposed to. Together with recent demonstrations of much higher spin Hall efficiences[39,40], the demonstrated locally embedded memristive tuning capability of individual SHNOs and their coupling in large arrays provide tremendous added functionality and a direct scaling path towards the realization of large nanoscale coupled oscillatory networks for a range of different low-power non-conventional computing models.

Gated SHNO and memristor geometries

After defining nano-constriction based SHNOs, to apply an electric field to the SHNO, we covered the entire W(5 nm)/$(Co_{0.75}Fe_{0.25})_{75}B_{25}$(1.7 nm)/MgO(2 nm)/$AlO_x$(2 nm)-based SHNO (see for example, refs.[23,36,37]) with 15 nm SiN_x and placed a 100-nm wide Ti/Cu gate on top of the 120-nm wide nano-constriction region. Figure 1a,b shows the schematic of the fabricated SHNO and the memristor in two operating regimes, HRS and LRS. The memristor shares its bottom electrode with the ground contact of the SHNO. The multi-level resistance states obtained by reversible memristive switching pinched at the origin is shown in Fig. 1c. The different LRSs of the memristor were achieved by sweeping the gate voltage while keeping the current compliance of the voltage source to different values; this controls the size of the conducting path formed inside the insulating multilayer (MgO(2 nm)/AlO_x(2 nm)/SiN_x(15 nm)). In practice, setting a memristor into various resistance states within an array of memristors is done by programming the width and amplitude of voltage pulses applied to the memristor electrodes[41,42]. The (negligible) memristor current $I_{\mathcal{M}}$ in the HRS is defined by the leakage current of the insulating multilayer. $I_{\mathcal{M}}$ in both the HRS and the LRS is added to or subtracted from the drive current applied to the SHNO (I_{SHNO}) depending on the memristor voltage polarity. Because of the presence of the metal-oxide layers and active upper electrode metals, MgO, AlO_x and Ti/Cu, the memristive switching mechanism is believed to be governed by either oxygen or metal ion migration forming a filamentary conduction bridge inside the insulating multilayer[30] as shown in Fig. 1a,b.

Figure 2 shows the memristive control of the power spectral density (PSD) for a 120-nm SHNO operating at $I_{SHNO} = 1.27$ mA while a magnetic field of $\mu_0 H = 0.3$ T was applied at an out-of-plane angle of $\theta = 65°$, chosen for an optimal amount of AO mode expansion to promote mutual synchronization[18,37,38], and an in-plane angle of $\phi = 22°$ for good anisotropic magnetoresistance read-out[18] (see Methods and Supplementary Fig. 5 for detailed description of the electrical measurements). The memristor is in the HRS when $V_{\mathcal{M}}$ is swept in the forward direction from −2 to 4 V as Fig. 2c shows a negligible value for $I_{\mathcal{M}}$. In the HRS the conductive path is not yet formed in the memristor, so the applied voltage induces a strong electric field across the insulating multilayer changing the PMA field ($H_k^{\perp}$) and consequently the effective magnetization described by $M_{eff} = M_S - H_k^{\perp}$. The change in M_{eff} directly translates into a change in the precession angle of the magnetization, resulting in an overall 60 MHz (10 MHz per V) AO frequency change as shown in Fig. 2a. When the memristor switches to one of its LRSs, in this case defined by a current compliance of 100 μA, at $V_{\mathcal{M}} = V_{set} = 4$ V, a non-negligible $I_{\mathcal{M}}$ is added to I_{SHNO} that in turn sharply increases the magnitude of spin current. The increase of the spin current is reflected in an experimentally observed jump in the SHNO frequency and each LRS defines a new frequency state (see Supplementary Fig. 1 for the detailed dependence). In the reverse sweep shown in Fig. 2b,d, the AO frequency linearly follows $I_{\mathcal{M}}$

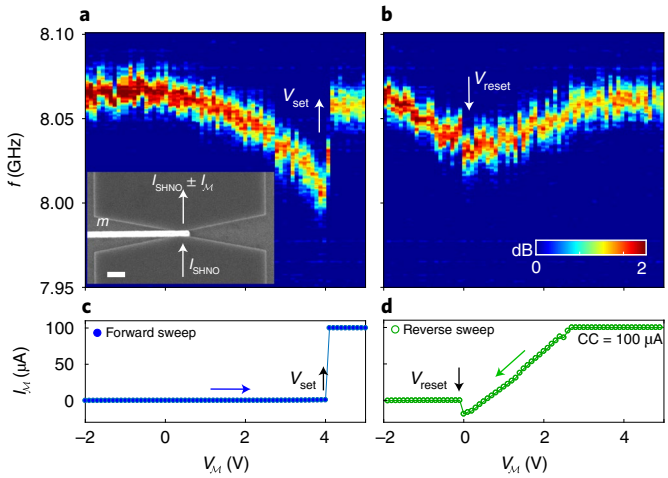

Fig. 2 | Memristive control of SHNO AOs. $I_{SHNO}=1.27$ mA in all plots. $\mu_0 H=0.3$ T was applied at $\theta=65°$ and $\phi=22°$. **a**, SHNO frequency tuned by forward sweep of the memristor voltage from its HRS, modifying the PMA by electric field until $V_{\mathcal{M}}=V_{set}$ where the AO is instead controlled by $I_{\mathcal{M}}$ in the LRS. Inset shows a scanning electron microscopy (SEM) image of the gate *m* on top of the SHNO. Scale bar, 300 nm. **b**, AO for the reverse $V_{\mathcal{M}}$ sweep. The memristor returns to its HRS at $V_{\mathcal{M}}=V_{reset}$ and the electric field regains control of the AO. **c**,**d**, The memristor currents, $I_{\mathcal{M}}$, for both forward (**c**) and reverse (**d**) sweeps.

until $V_{\mathcal{M}}$ reaches V_{reset} and $I_{\mathcal{M}}$ again becomes negligible. When the SHNO chain is biased, the memristor *IV* plot is no longer pinched at 0 V since the bottom electrode is at a non-zero potential. From V_{reset}, the memristor is back to its HRS where the AO frequency is controlled by the electric field. One fundamental importance of the memristive gating hence lies in the fact that the memristive mechanism provides two qualitatively very different regimes with their respective tuning knobs for the frequency of the SHNO: an electric field and a memristor current.

To study the potential for using both the electric field and the memristor current to tune individual nano-constrictions in SHNO arrays, we fabricated a chain accommodating two SHNOs and placed two memristive gates, $\mathcal{M}_1$ and $\mathcal{M}_2$, on top of the nano-constriction regions as shown in the inset in Fig. 3a. When both gates are floating, that is, no voltage applied to either $\mathcal{M}_1$ or $\mathcal{M}_2$, sweeping I_{SHNO} brings the two SHNOs into synchronization at about $I_{SHNO}=0.8$ mA as shown in Fig. 3a. The SHNOs stay synchronized until I_{SHNO} reaches 0.9 mA. As our initial state, we therefore fix I_{SHNO} to 0.85 mA corresponding to a stable synchronized state.

Figure 3b shows the memristive switching of $\mathcal{M}_1$ as a function of the applied gate voltage, $V_{\mathcal{M}_1}$. Starting from $V_{\mathcal{M}_1}=-2$ V, $\mathcal{M}_1$ stays in its HRS until it switches to its LRS at $V_{set}=2.4$ V (blue solid data points and arrow). On reversing the voltage sweep direction, $\mathcal{M}_1$ stays in its LRS until it is reset at about −0.2 V (green hollow data points and arrow). The corresponding response of the SHNO chain is shown in Fig. 3c. We first note that applying a negative voltage to $\mathcal{M}_1$ breaks the synchronized state into two distinct frequency branches, f_1 and f_2, with lower peak power and larger linewidth (Δf) values as shown by the red and blue solid data points in Fig. 3d. As $V_{\mathcal{M}_1}$ approaches zero, the synchronized state is retrieved and appears to grow more robust since both peak power and Δf continues to improve. However, once $\mathcal{M}_1$ switches to its LRS, a substantial $I_{\mathcal{M}_1}=100$ μA ($I_{\mathcal{M}_1}>0$) is sharply injected into the entire chain causing a jump in the operating frequency, as shown by the white arrow in Fig. 3a. At this operating point, the chain is no longer robustly synchronized and both peak power and Δf deteriorate, as shown by hollow green data points in Fig. 3d. However, as Fig. 3e depicts, when $I_{\mathcal{M}_1}$ is reduced during the reverse sweep, the synchronized state is once again recovered even when $I_{\mathcal{M}_1}$ eventually changes its sign ($I_{\mathcal{M}_1}<0$) and opposes I_{SHNO} while $V_{\mathcal{M}_1}$ crosses the pinch off voltage. Finally, the synchronization again breaks when $\mathcal{M}_1$ is back to the HRS at $V_{\mathcal{M}_1}=V_{reset}$. A very similar behaviour was observed when forward and reversed sweeps of $V_{\mathcal{M}_2}$ were performed on $\mathcal{M}_2$ (Supplementary Fig. 2).

We then studied the electric field and memristive tuning within a chain accommodating four SHNOs in series. Two memristive gates, $\mathcal{M}_1$ and $\mathcal{M}_2$, were again fabricated but this time we attempted to place them on top of the bridges connecting the SHNOs 2 and 3 and 3 and 4 as shown in the inset in Fig. 4a. As alignment of multiple e-beam exposures at the nanoscale is particularly difficult, in the final device the gates are positioned approximately halfway between the constriction and the bridge (Fig. 4a), which is the device that will be studied in detail below.

The PSD profile of the chain versus I_{SHNO} (both gates floating), shown in Fig. 4a, indicates that the four oscillators start in a synchronized state and later break apart into two frequency branches for larger I_{SHNO}. Since the power and linewidth of the two branches are similar to each other, it is reasonable to assume that they correspond to pair-wise mutual synchronization; the lower frequency branch then disappears, which we interpret as it breaking up into individual oscillators with signals too weak to be observed. Since the coupling is primarily nearest-neighbour, one branch should correspond to SHNOs 1 and 2 and the other to SHNOs 3 and 4. As will become clear from the effect of gating shown in Fig. 4c–g, and from the results discussed later in Fig. 5, the lower frequency branch corresponds to SHNOs 1 and 2, and the upper to SHNOs 3 and 4. We therefore label the two branches $f_{1,2}$ and $f_{3,4}$, respectively, in Fig. 4.

A value of $I_{SHNO}=0.712$ mA (dotted white arrow in Fig. 4a), at which the synchronization is broken into the two partially synchronized branches, was chosen as the operating point to conduct the subsequent gating operation. In the HRS (indicated by the small $I_{\mathcal{M}_1}$, Fig. 4b), sweeping $V_{\mathcal{M}_1}$ forward (while $\mathcal{M}_2$ is floating) brings the upper frequency branch down while the lower frequency branch remains largely unaffected, as shown in Fig. 4c. The behaviour of the upper branch is reminiscent of that seen earlier in the single constriction in Fig. 2a. As the gates only change the local electric field, we hence conclude that the immediate impact of gating on the upper frequency branch, while the corresponding relative lack thereof on the lower frequency branch, both indicate that the lower frequency branch corresponds to SHNOs 1 and 2 and the upper branch to SHNOs 3 and 4. We can also conclude that gates positioned halfway between the constrictions and the bridge still affect the intrinsic frequencies strongly.

At about $V_{\mathcal{M}_1}=1.5$ V, the upper branch bends more sharply downwards and the linewidths of both branches deteriorate substantially; this indicates the onset of pronounced mutual interaction between the two branches. While this may indicate that the higher gate voltage increases the interoscillator coupling, consistent with their impact on the connecting bridges, it could also be due to frequency pulling or a combination of these independent effects. At a slightly higher $V_{\mathcal{M}_1}=V_{sync}$ (white arrow in Fig. 4c) the two branches then merge and continue to operate in unison as also indicated by the 50% drop of the linewidth and the substantial increase in the peak power (Supplementary Fig. 3); the chain remains synchronized even when $\mathcal{M}_1$ switches to the LRS and at 4.2 V and adds $I_{\mathcal{M}_1}=100$ μA to SHNOs 3 and 4. In the HRS, when $V_{\mathcal{M}_1}$ is swept from −3 to +4.2 V, an overall frequency change of 200 MHz is achieved, corresponding to a tunability of 28 MHz per V.

For the reverse voltage sweep direction presented in Fig. 4d, the synchronization state is extended for a larger range of $V_{\mathcal{M}_1}$ down to $V_{un-sync}$ (white arrow in Fig. 4d) as the oscillators tend to pull each other's frequencies when they are synchronized, a phenomenon known as hysteretic synchronization[43]. To conclude, $\mathcal{M}_1$ operating

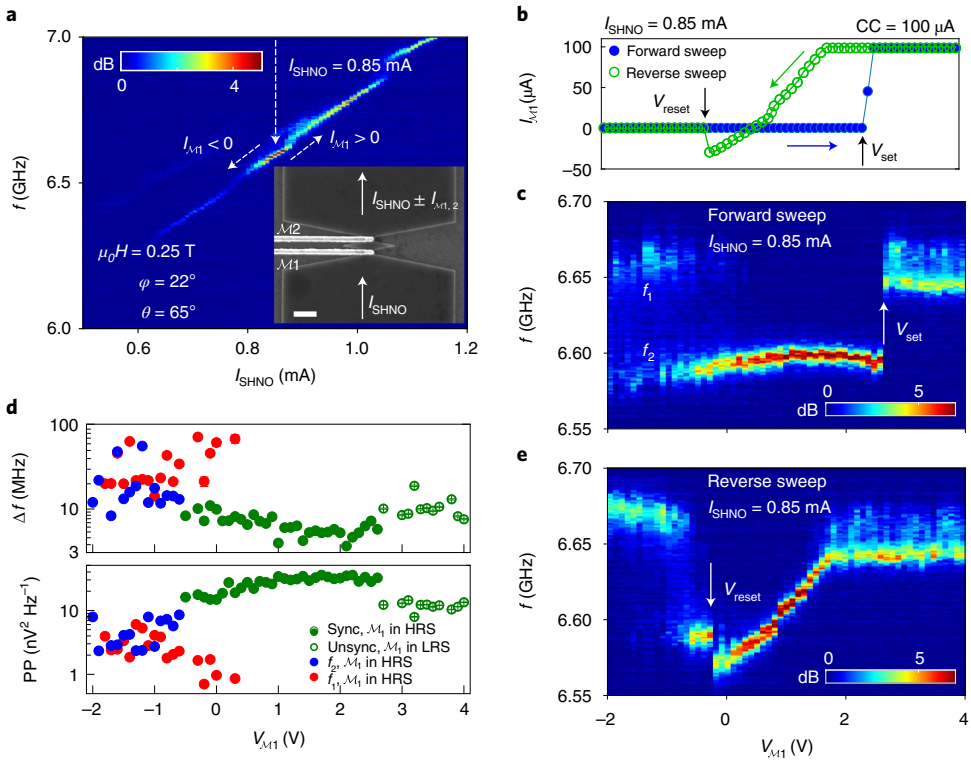

Fig. 3 | A chain of two SHNOs with two memristive gates. a, PSD versus I_{SHNO} in a field of $\mu_0 H = 0.25$ T applied at $\theta = 65°$ and $\phi = 22°$. The AO shows a synchronized region for a small I_{SHNO} range around 0.85 mA. The inset shows an SEM image of the two nano-constrictions with their respective memristor gates, $\mathcal{M}_1$ and $\mathcal{M}_2$. Scale bar, 300 nm. **b**, $I_{\mathcal{M}_1}$ versus $V_{\mathcal{M}_1}$ showing reversible switching between HRS and LRS defined by CC = 100 μA for a constant $I_{SHNO} = 0.85$ mA. **c**, PSD versus $V_{\mathcal{M}_1}$ for the forward sweep. The electric field induced by $V_{\mathcal{M}_1}$ in the HRS breaks the synchronized state and recovers it again as $\mathcal{M}_1$ approaches $V_{\mathcal{M}_1} = V_{set}$. In the LRS, an $I_{\mathcal{M}_1} = +100$ μA adds to the I_{SHNO} increasing the AO to a non-synchronized frequency in **a**. **d**, Extracted peak power and linewidth values versus $V_{\mathcal{M}_1}$ for the forward sweep in **c**. **e**, PSD versus $V_{\mathcal{M}_1}$ for reverse sweep. The AO frequency linearly follows $I_{\mathcal{M}_1}$ recovering the synchronized state until $\mathcal{M}_1$ switches back to its HRS and synchronization breaks apart again.

in the HRS is capable of either bringing multiple oscillators to a synchronized state or break their synchronization apart. The maximum power consumption of the tuning provided by $I_{\mathcal{M}_1}$ in the HRS is only 0.4 μW, which can be further suppressed by material engineering for the insulating multilayer.

Turning now to the impact of the second gate, the memristive switching of $\mathcal{M}_2$ is first shown in Fig. 4e, with a behaviour that is reasonably similar to that of $\mathcal{M}_1$. We then show the impact of sweeping $V_{\mathcal{M}_2}$ in the forward direction in Fig. 4f. At negative voltages we observe one strong signal and another that is barely visible. Noting that the strong signal has a higher power than the pair-wise synchronized states in Fig. 4c,e, and the weak signal has one that is lower, it is plausible to conclude that the strong signal is a result of three mutually synchronized oscillators, while the weak signal is a lone oscillator. The most likely scenario is then that the strong signal comes from a mutually synchronized state of oscillators 1, 2 and 3, while the weak signal is oscillator 4. We have therefore labelled the two branches $f_{1,2,3}$ and f_4 in this region.

It is possible to reconcile the different effects of $\mathcal{M}_1$ and $\mathcal{M}_2$ gating if one posits that the free-running frequency of oscillator 3 is in between those of oscillators 1 and 2 on the one hand and oscillator 4 on the other. As known from Fig. 4a, without any applied voltage, oscillator 3 will mutually synchronize with oscillator 4. However, when a negative voltage is applied to $\mathcal{M}_2$ in Fig. 4f,g, the frequency of oscillator 4 is increased too far away from that of oscillator 3, which allows the latter to instead synchronize with oscillator 1 and 2. This is different from when a voltage is applied to $\mathcal{M}_1$, as this primarily increases the frequency of oscillator 3, which makes the synchronization to oscillator 4 more robust. As $V_{\mathcal{M}_2}$ approaches zero, both f_4 and $f_{1,2,3}$ decrease, the former due to a change in its PMA, and the latter probably from interaction (frequency pulling) with f_4. This 3/1 partially synchronized state persists up until about 0.5 V, where it changes into pair-wise 2/2 synchronization as the frequencies of oscillators 3 and 4 are again close enough to instead favour this arrangement. When $\mathcal{M}_2$ switches into its LRS at about 3 V, the upper branch experiences a larger frequency step than the fully synchronized step in Fig. 4c did, and the lower branch experiences a smaller one (the difference is almost a factor of two); this behaviour is again consistent with a direct memristor current influence on oscillators 3 and 4 and frequency pulling on oscillators 1 and 2.

Reversing the $\mathcal{M}_2$ voltage sweep direction follows the same general behaviour (Fig. 4g): the pair-wise synchronized state shows a stronger voltage dependence of the directly effected upper branch compared to the weaker dependence of the frequency pulled lower branch. This state persists down to about −0.7 V, where oscillator 3 breaks away from oscillator 4 and instead joins the lower branch. Using the gates, it is hence possible to switch between three different mutually synchronized states: 2/2, 3/1 and fully synchronized.

Finally, we show how the memristor current itself can dominate over the VCMA effect and hence promote synchronization, or break it reversibly. Since we have found that we can provide stronger memristor currents in the forward current direction than in the negative, we now operate the same chain as in Fig. 4 but with the direction of the applied field and current reversed. The positive direction of I_{SHNO} is shown by white solid arrows in the inset of Fig. 5a. Similar to Fig. 4a, the chain starts in synchrony but fails to hold its state and splits into two frequency branches, $f_{1,2}$ and $f_{3,4}$, at more negative I_{SHNO}. Having fixed I_{SHNO} at −0.757 mA, we swept $V_{\mathcal{M}_1}$

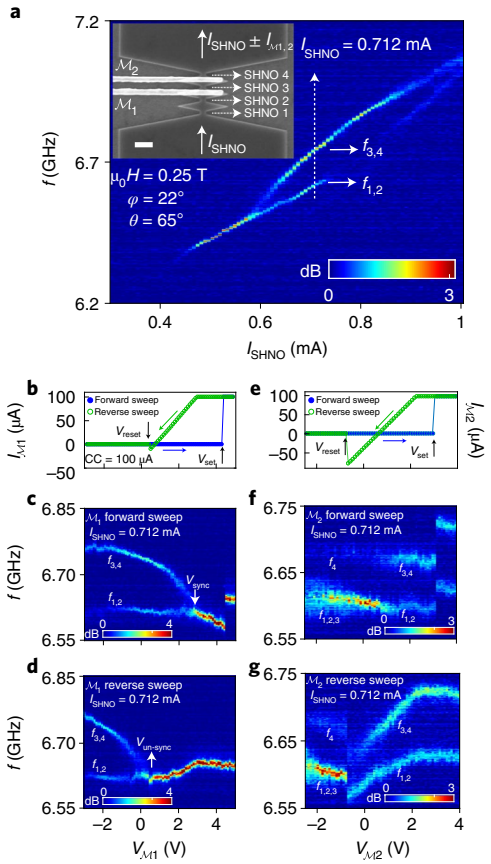

Fig. 4 | VCMA dominant control of synchronization in a chain of four SHNOs with two memristive gates. a, PSD versus I_{SHNO} in a field of $\mu_0 H = 0.25$ T applied at $\theta = 65°$ and $\phi = 22°$. The AO starts in a synchronized state, then splits into two branches at higher I_{SHNO}. The inset shows an SEM image of the SHNO with two memristor gates, $\mathcal{M}_1$ and $\mathcal{M}_2$, placed mid-way between the nano-constrictions and the bridges connecting the SHNOs. For $I_{SHNO} = 0.712$ mA. Scale bar, 300 nm. **b**, $I_{\mathcal{M}_1}$ versus $V_{\mathcal{M}_1}$ shows the memristive switching of $\mathcal{M}_1$. **c**, PSD versus $V_{\mathcal{M}_1}$ in the forward sweep where the two frequency branches synchronize by VCMA in the HRS. The chain remains synchronized even when $\mathcal{M}_1$ switches to the LRS. **d**, PSD response for reverse sweep of $V_{\mathcal{M}_1}$. The chain stays in synchrony until $\mathcal{M}_1$ toggles to the HRS again. **e**–**g**, $I_{\mathcal{M}_1}$ versus $V_{\mathcal{M}_2}$ and the memristive switching of $\mathcal{M}_2$ having very weak impact on recovering the broken synchronized state (**e**) in neither the forward (**f**) nor the reverse (**g**) sweeps of $V_{\mathcal{M}_2}$.

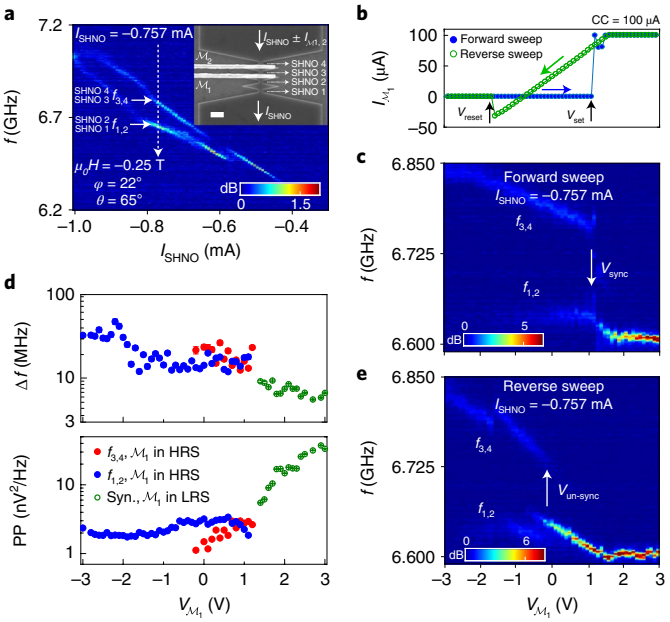

Fig. 5 | Memristive dominant control of synchronization in a chain of four SHNOs with two memristive gates. a, PSD versus I_{SHNO} in a reversed field of $\mu_0 H = -0.25$ T applied at $\theta = 65°$ and $\phi = 22°$. The AO starts in a weak synchronized state, then again splits into two branches at more negative I_{SHNO}. The arrows in the inset define the positive direction of I_{SHNO}. Scale bar, 300 nm. **b**, $I_{\mathcal{M}_1}$ versus $V_{\mathcal{M}_1}$ showing the memristive switching of $\mathcal{M}_1$. **c**, PSD versus increasing $V_{\mathcal{M}_1}$ at $I_{SHNO} = -0.757$ mA. While the VCMA in the HRS is insufficient to synchronize the SHNOs, the substantial $I_{\mathcal{M}_1}$ added when $\mathcal{M}_1$ switches to its LRS, sharply pulls the $f_{3,4}$ branch towards $f_{1,2}$ and synchronizes the entire chain. **d**, shows how the corresponding extracted linewidth and peak power (PP) values improvement when $\mathcal{M}_1$ switches. **e**, PSD versus decreasing $V_{\mathcal{M}_1}$. The synchronized state is first tuned by the decreasing $|I_{\mathcal{M}_1}|$ and then breaks apart when $V_{\mathcal{M}_1}$ switches at V_{reset}.

in both the forward and reverse directions and obtained a robust memristive switching as shown in Fig. 5b. Note that due to the negative sign of I_{SHNO}, the memristor IV is now pinched at a negative $V_{\mathcal{M}_1}$ value. Figure 5c shows that during the forward sweep, while the VCMA $V_{\mathcal{M}_1}$ bends the upper frequency branch down, it fails to place it within the synchronization bandwidth with the lower frequency branch. This is in contrast to Fig. 4c where VCMA alone can make the two frequency branches synchronize. However, when $\mathcal{M}_1$ toggles to its LRS at $V_{\mathcal{M}_1} = V_{set} = V_{synch}$, a noticeable $I_{\mathcal{M}_1}$ flows to the common ground contact through SHNO 3 and 4, this time opposing I_{SHNO}. Therefore, the operating current for SHNO 3 and 4 drops sharply, which is reflected as a sharp decline in the upper frequency branch. Since $I_{\mathcal{M}_1}$ only affects SHNO 3 and 4, we can again, as in Fig. 4, associate the operating frequency of SHNO 3 and 4 with the upper frequency branch, hence again indicated as $f_{3,4}$. As a result of the sharp frequency jump, the $f_{3,4}$ branch jumps down to a frequency value near $f_{1,2}$ where they synchronize as shown in Fig. 5d indicating substantial improvement in the peak power and linewidth values (hollow green data points). Reversibly, in Fig. 5e, when $|I_{\mathcal{M}_1}|$ is too small at $V_{\mathcal{M}_1} = V_{un-sync}$ the synchronized state breaks although $\mathcal{M}_1$ is still in the LRS. The two frequency branches keep increasing their distance when V_{reset} is met and electric field takes the control back.

The impact of adding current through $\mathcal{M}_2$ is shown in Supplementary Fig. 4. A fully synchronized state can no longer be reached. Instead, the partially synchronized state can be switched between 3/1 (high positive voltage), 2/2 (around zero voltage) and possibly also 1/3 (strong negative voltage).

Prospect for non-conventional computing and suggested implementation

Compared to tuning applied magnetic field and direct current (d.c.) to control and improve coherency and output power of SHNO arrays as microwave sources, the gating approach offers tunability at a much lower power and anywhere within SHNO arrays at nanoscale. Therefore, apart from improving coherency and output power of SHNO array, the low-power gating also holds great potential as emerging computing systems. All coupled oscillatory network-based computing schemes, irrespective of their topology and hardware implementation require a certain input interface, for example, to provide a bus to feed the data in pattern recognition[44,45], map the problem to hardware[46,47], adjusting the weights in neural networks (for example, deep neural networks), training the output layer in reservoir computing and tune a reservoir to operate optimally at the edge of chaos[48]. The last two points are missing pieces in

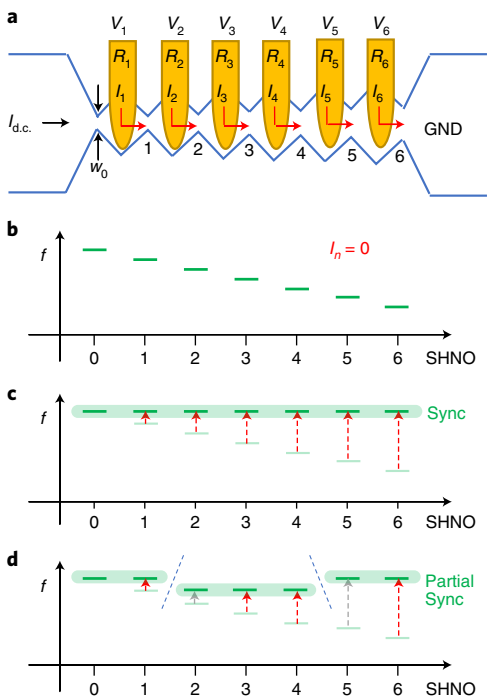

Fig. 6 | Pattern matching using a memristive SHNO chain. a, The SHNO chain consists of seven nano-constrictions (0–6) connected by six bridges. A global operating current ($I_{d.c.}$) is pushed to ground (GND) through the chain such that all nano-constrictions operate in a regime where they interact and their frequency increases linearly with current. The width (w_0) of the first constriction is indicated; the widths of all subsequent nano-constrictions are slightly wider than their upstream neigbor such that its frequency is slightly lower. On top of each bridge is a memristive gate, through which additional current can be added locally to the global current. **b**, AO frequency versus nano-constriction number when no additional current is provided by the gates. **c**, AO frequency of the same SHNO chain when each gate voltage matches its programmed resistance such that the frequency of each nano-constriction reaches that of its upstream neighbour. As a consequence, all SHNOs mutually synchronize on the frequency of the first SHNO. **d**, AO frequency of the same chain when two of the input voltages do not match their programmed resistances (indicated by grey arrows). Here, V_2 is too low, and V_5 is too high and as a consequence the chain is only partially synchronized as it breaks up into three different segments (cuts indicated by blue dashed lines).

recent reservoir computing proposals[22,49] for large nanoscale oscillator arrays.

Finally, we want to present one such example of how a memristor controlled SHNO chain may be used for programmable ultrafast pattern recognition. The memristors will be operated in their LRS where the resistance of each gate can be tuned between 15 and 50 kΩ through the process shown originally in Fig. 1; the memristor resistance is about 10–100× higher than the resistance of the nano-constriction, $R_{SHNO} \cong 200\,\Omega$. When a voltage V_m is applied to one such gate, its SHNO and all other SHNOs further downstream, receive an additional current $I_m = V_m/R_m$, which is a small fraction of the global drive current $I_{d.c.}$. As the SHNO frequency increases with current, the added I_m results in a slight increase in the frequency of the SHNO underneath the gate and all subsequent SHNOs. As we want to use the different I_m for matching, we will design the SHNO chain such that it is far from its mutually synchronized state for $I_m = 0$ (that is, without any voltage inputs on the gates) but can reach mutual synchronization if typical gate currents are added. We can achieve this functionality by making each subsequent nano-constriction slightly wider than its upstream neighbour (Fig. 6a) such that each nano-constriction auto-oscillates on a frequency f_n a $\Delta f = f_n - f_{n-1}$ lower than that of its upstream neighbour, where Δf is outside the locking bandwidth for mutual synchronization (Fig. 6b). To reach a state of complete mutual synchronization (Fig. 6c), each SHNO needs a slightly higher current than its upstream neighbour. This additional current will then be provided by each gate. Through the tunable R_m of each gate a particular voltage input vector will then be required for the chain to reach mutual synchronization. Figure 6d shows what happens if not all input voltages match their programmed resistances. In this specific case, the second voltage is too low and the fifth voltage is too high, while all other voltages match their respective programmed values. The poor match of two of the voltages breaks the mutual synchronization of the SHNO chain at these two locations and the chain breaks up into three independently synchronized segments. As the output power grows as the square of the number of mutually synchronized SHNOs, this partially synchronized state can then be readily distinguished from a fully synchronized state. Micromagnetic simulations of a chain with three constrictions and two memristive gates are shown in the Supplementary Fig. 6 and confirm the principle behind this particular implementation.

Outlook

Memristive control of SHNOs provides a scalable approach to both direct input and non-volatile reversible fine-tuning of the individual oscillators and their connections at any location within the network. The memristive approach not only offers such tunability at extremely low power but also opens up a wide range of opportunities to use application-specific memristors from the entire family of memristors[30,50] depending on the computing model for which the oscillator array is designed. More importantly, the memristive gating approach inherently addresses the memory bottleneck from which conventional von Neumann processors suffer. In memristor controlled arrays, each processing unit, that is, each SHNO, has its own memory element and there is hence no need to store and retrieve their states in any external memory limited by data transfer bandwidth. This in turn makes such oscillatory networks with intrinsic memory appealing candidates for edge computing models that aim to bring computation and data storage closer to the location where they are needed.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41563-021-01153-6.

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Methods

Fabrication of memristive controlled SHNO. A trilayer stack of W(5)/$(Co_{0.75}Fe_{0.25})_{75}B_{25}$(1.7)/MgO(2) (thicknesses in nm) was grown at room temperature on an intrinsic high resistivity Si substrate ($\rho_{Si} > 10\,k\Omega\,cm$) using an ultra-high vacuum sputtering system. Direct current and radio frequency sputtering were sequentially used for the depositions of metallic and insulting layers, respectively. The stack was covered with 2-nm thin layer of sputtered AlO_x at room temperature to protect the MgO layer from degradation due to exposure to the ambient conditions. The stack was subsequently annealed at 300 °C for 1 h to induce PMA. The stack was then patterned into an array of $4 \times 14\,\mu m^2$ rectangular mesas and the nano-constriction SHNO devices with different widths were defined at the centre of these mesas by a combination of electron-beam lithography and argon ion beam etching using negative electron-beam resist as the etching mask. After removing the electron beam resist, the sample was covered with 15 nm stoichiometric room temperature sputtered SiN_x to isolate the gate contacts from the nano-constriction metallic sidewalls. The gates were defined by sputtering a bilayer of Ti(2 nm)/Cu(40 nm) followed by electron-beam lithography using negative electron resist to fabricate 100-nm wide gate pattern on top of nano-constriction. The pattern was then transferred to the Ti/Cu bilayer using an ion beam etching technique. After removing the remaining negative resist, the sample went through optical lithography using positive resist to define vias in SiN_x, giving access to the SHNO metal layers for the contact pads. Finally, the electrical contacts, including two SG-CPWs for microwave and d.c. measurements were defined by optical lithography and lift-off for a bilayer of Cu(700 nm)/Pt(20 nm).

Memristor characterization and microwave measurements. All microwave electrical measurements were carried out at room temperature using a custom-built probe station with the sample mounted at a fixed in-plane angle on an out-of-plane rotatable sample holder between the pole pieces of an electromagnet capable of producing a uniform magnetic field. Using a 6220 Keithley current source, a direct positive electric current, I_{SHNO}, was made to inject through the d.c. port of a high frequency bias-T and the resulting auto-oscillating signal was then amplified by a low-noise amplifier with a gain of +46 dB and subsequently recorded using a spectrum analyser from Rhode and Schwarz (10 Hz–40 GHz) comprising a resolution bandwidth of 1 MHz. Two 2400 Keithley source meters were used to sweep the voltage to perform the voltage sweeps on the memristors. The output current of the source meter was compiled by setting different values for the current compliance on the instrument. Also, the source meter measured the current of the memristive gates for any applied voltage confirm the proper switching between HRS and LRSs (see Supplementary Fig. 5 for schematic of the measurement setup).

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author on request.

Code availability

The MATLAB and MuMax codes used in this study are available from the corresponding author on reasonable request.

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Author contributions

S.F., S.K. and H.O. developed the material stacks. M.Z. designed and fabricated the devices, and carried out all measurements and data analysis. H.F. and A.H contributed to the data analysis. R.K. and M.D. carried out all micromagnetic simulations. J.Å. led the project. All authors contributed to the interpretation of the results and cowrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Johan Åkerman.

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